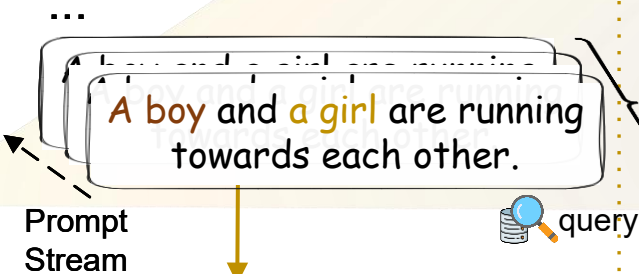


LLM Layout Planning

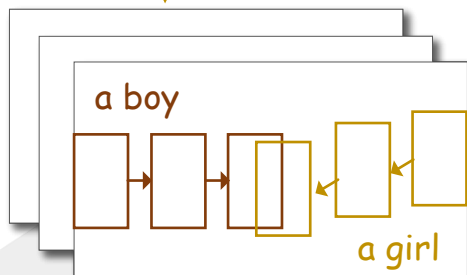


You are an intelligent spatiotemporal layout planner for videos. Generate bounding boxes (Bbox) for visual objects.

- Bbox rules: prompt following, consistency
- Output format: Frame i : {'id': <int>, 'name': <object name>, 'box': [x, y, width, height]} ... (In-context examples)

Layout Generation + Consistency Check

$\{Y_0, \dots, Y_N\}$



Spatiotemporal Layout

Text Encoder

denoising

denoising

DiT

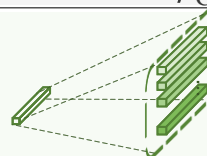
read & load

unload & insert

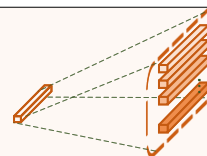
Parametric Memory

Abstraction of C $g(C)$ ϕ_C^*

<obj A> drifts right to left above <obj B>.

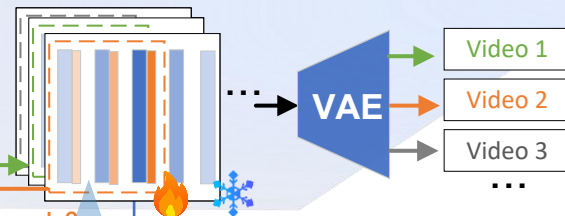


Two <objects> runs towards each other.



$$\mathcal{M} = \{g(C) : \phi_C^* \mid C \in \mathcal{C}_H\}$$

Test-Time Optimization



Attention-Layout Relevancy Probe

Cross-Att.

Object-level Att.

Layer i

Layer i

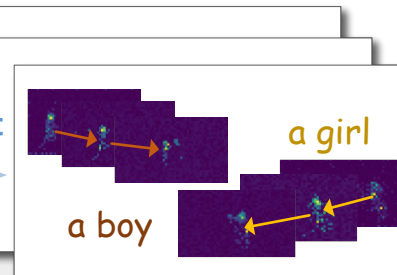
Layer j

Layer j

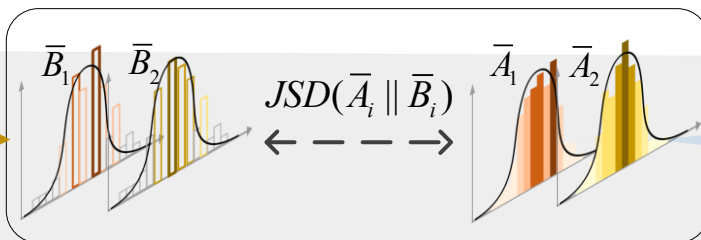
a boy a girl

$$\bar{A}_i = f_{\theta}(C, c_i, z_i, t)$$

Gradient



Cross-Attention Maps



Attention-to-Layout Alignment